

Mithilessh Bhasker

STUDENT DEVELOPER • FTC ROBOTICS SOFTWARE LEAD

Software Lead

FTC Team 16481

12 Merged PRs

Pedro Pathing

Kumon Instructor

2 years

Kumon Reading Program

Completer

EDUCATION

Amador Valley High School

Pleasanton, California

Expected Graduation: 2029

SKILLS

- Java
- Python
- JavaScript / TypeScript
- React / Next.js
- Git & GitHub
- FTC SDK / Pedro Pathing
- KiCad / PCB Design
- Onshape / Prototyping

CERTIFICATES

CS50x

Harvard University

Cybersecurity Fundamentals

IBM SkillsBuild

Fundamentals of Remote Sensing

NASA ARSET

Earth Observations for Reservoir and Water Utility Management

NASA ARSET

Stardance Challenge

Hack Club

RECOGNITION

- Control Award - NorCal Regional Championship
- Sustain Award - 3rd Place - Western Edge
- Enrico Fermi Discovery Award - 2nd Place - CRI
- Connect Award - Folsom Qualifier
- Inspire Award - 2nd Place + Finalist Alliance Captain
- Think Award - FIRST Tech Challenge
- AoPS Algebra 2 - 1st Place
- AoPS High School Contest Math - 1st Place
- Kumon Reading Program - Full Completion

EXPERIENCE / LEADERSHIP

• Robo Racers — Software Lead

FTC Team 16481

Lead autonomous software development for the competition robot, turning match strategy into repeatable field performance.

- Build and tune autonomous routes around scoring positions, obstacles, timing, and robot orientation.
- Use localization, pose, heading, path state, and telemetry to diagnose failures and improve reliability.

• Pedro Pathing — Contributor + Beta Tester

12 merged PRs

Improve the Quickstart and documentation used by FTC teams to tune, localize, and build autonomous routines.

- 4 Quickstart changes across two- and three-wheel tuners, auto-tuning, and configuration.
- 8 documentation updates covering localization, coordinates, Bézier curves, constraints, and setup.

• Kumon — Instructor

2 years

Worked as a Kumon Instructor, supporting independent learning, explaining concepts, checking work, and encouraging consistent progress.

- Completed the entire Kumon Reading Program, building advanced comprehension, vocabulary, analysis, discipline, and time-management skills.

• Equal Horizons — Co-founder

Assistive technology

Co-founded accessible assistive-tech product and web prototypes.

SELECTED PROJECTS

Jarvis Voice Assistant

Python • local processing

Custom voice assistant with modular commands and local-first integrations.

OmniSight HUD

Computer vision • robotics

Heads-up display concept for real-time robotics information and clean visual feedback.

Stream Deck PCB

KiCad • hardware

Custom macro pad designed from scratch with programmable keys and a clean PCB layout.

Mycroft Mark II Dev Board

KiCad • embedded systems

Hardware development and board-level prototyping inspired by production PCB systems.